

Theory Exercises

1.1

(4b) We observe that $K_4(s)$ exhibits an oscillatory behaviour, in which the curvature oscillates as a sine but with increasing amplitude as s increases. This results in larger variations in the curve's turning rate at higher values of s , which matches with the observed loops in b

(3d) $K_3(s) = s^3 - 4s = s(s^2 - 4) = s(s-2)(s+2)$ So the curvature changes sign at $s=0$ and symmetrically at $s=\pm 2$, meaning the curvature switches bending direction at these points, with opposite curvatures on either side. Corresponds to figure d

(1a) we observe that $\frac{s^2-1}{s^2+1} = 0 \rightarrow s^2-1=0 \rightarrow s=\pm 1$ Two inflection points where the curvature sign changes. we also have that

$$\lim_{s \rightarrow \pm \infty} K_1(s) = \lim_{s \rightarrow \pm \infty} \frac{s^2-1}{s^2+1} = 1 \quad \text{The curve will resemble a circle of radius 1 for large values of } s \text{ as we see in a}$$

(2c) The function is symmetric around the origin as we expect from $K_2(s)=s$, where curvature changes sign at $s=0$. We also observe that $\lim_{s \rightarrow \pm \infty} K_2(s) = \pm \infty$ meaning that as s grows the curvature grows indefinitely and then the radius of curvature will go to zero as we see on image c

1.2

We define the dome in spherical coordinates where $R=10m$, θ is the azimuth angle that goes from 0 to 2π and φ is the polar angle going from 0 to $\frac{\pi}{2}$. The surface element for a sphere is:

$$dA = R^2 \sin(\varphi) d\varphi d\theta$$

And we also know that height can be expressed as:

$$z = R \cos(\varphi) = 10 \cos(\varphi)$$

So we can see that the total area is

$$\begin{aligned} \text{PART B} \rightarrow A &= \left| \int_0^{2\pi} \int_{\varphi_1}^{\varphi_2} 100 \sin(\varphi) d\varphi d\theta \right| = \quad \text{where } \varphi_1 = \arccos\left(\frac{0}{10}\right) = \frac{\pi}{2} \\ &= \left| \int_0^{2\pi} \int_{\frac{\pi}{2}}^{\frac{\pi}{3}} 100 \sin(\varphi) d\varphi d\theta \right| = \left| 200\pi \cdot \left[-\cos(\varphi) \right]_{\frac{\pi}{2}}^{\frac{\pi}{3}} \right| = \left| 200\pi \cdot \left(-\frac{1}{2} - 0 \right) \right| = 100\pi \end{aligned}$$

$$\begin{aligned} \text{PART A} \rightarrow A &= \left| \int_0^{2\pi} \int_{\varphi_1}^{\varphi_2} 100 \sin(\varphi) d\varphi d\theta \right| = \quad \varphi_1 = \arccos\left(\frac{5}{10}\right) = \frac{\pi}{3} \\ &= \left| 200\pi \cdot \left[-\cos(\varphi) \right]_{\frac{\pi}{3}}^0 \right| = 100\pi \quad \varphi_2 = \arccos\left(\frac{10}{10}\right) = 0 \end{aligned}$$

The area of both parts is equal so there would not be any difference in the case of two parts. let's see now with N slices.

The K -th slice spans from $z_K = R - K \cdot \frac{R}{N}$ to $z_{K+1} = R - (K+1) \cdot \frac{R}{N}$ so $\varphi_K = 1 - \frac{K}{N}$ and $\varphi_{K+1} = 1 - \frac{K+1}{N}$. The total area of a slice is then:

$$A = \int_0^{2\pi} \int_{1-\frac{K+1}{N}}^{1-\frac{K}{N}} R^2 \sin(\varphi) d\varphi d\theta = 2\pi R^2 \cdot \left(\cos\left(1-\frac{K+1}{N}\right) - \cos\left(1-\frac{K}{N}\right) \right)$$

We can use a Taylor expansion of the cosine $\cos(x + \Delta x) \approx \cos(x) - \Delta x \sin(x)$

$$A \approx 2\pi R^2 \left[\cos\left(1-\frac{K}{N}\right) + \frac{1}{N} \cdot \left(-\sin\left(1-\frac{K}{N}\right)\right) - \cos\left(1-\frac{K}{N}\right) \right] = 2\pi R^2 \left[-\frac{1}{N} \sin\left(1-\frac{K}{N}\right) \right]$$

THE HIGHEST K WILL HAVE THE BIGGEST PROFIT → YOU WANT THE HIGHEST PART OF THE DOME

DECREASES AS K INCREASES